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Regression Discontinuity and the Price Effects of Stock Market Indexing

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1 Big picture

- **Question.** What is the causal price effect of stock-market indexing? When a stock is added to or deleted from a major index, does forced buying or selling move prices, and by how much?
- **Setting.** The paper studies the annual reconstitution of the Russell 1000 and Russell 2000 indexes from 1996–2012. Stocks are ranked by end-of-May market capitalization. Stocks ranked 1–1000 go into the Russell 1000; stocks ranked 1001–3000 go into the Russell 2000.
- **Core challenge.** Traditional event studies of index changes, especially for the S&P 500, are contaminated by news, attention, or investor-recognition effects. It is hard to know whether price changes come from indexing demand or from other information released around index inclusion.
- **Main empirical idea.** The Russell reconstitution creates a regression-discontinuity setting. Stocks just below and just above the 1000 cutoff are very similar in size, but their index weights differ sharply after reconstitution because the Russell 2000 is value-weighted and the top of that index receives much more weight than the bottom of the Russell 1000.
- **Why the design is powerful.** Around the cutoff, small random differences in end-of-May market capitalization create large discontinuous changes in index demand. This gives a clean quasi-experiment for the price effect of indexing.
- **Main return results.**
 - Additions from the Russell 1000 into the Russell 2000 earn about **5.0% higher June returns**.
 - Deletions from the Russell 2000 into the Russell 1000 experience about **5.4% lower June returns**.
 - These effects are concentrated in **June**, the reconstitution month, with little evidence of pre-trends or later reversals.
- **Main interpretation.** The symmetry between additions and deletions supports a *forced-buying / forced-selling* interpretation rather than an investor-recognition story.
- **Elasticity result.** Using the benchmarked-demand shock implied by the discontinuity, the paper estimates a stock-demand elasticity of about -1.46 using total benchmarked assets, or about -0.39 using passive assets only.
- **Time-trend result.** Even though more money became indexed over time, the price impact of indexing *fell*. At the same time, trading-volume effects grew. This suggests that arbitrage and liquidity provision became more effective over time.
- **Liquidity-provider result.** Funds that provide liquidity to Russell 2000 indexers tend to hold **larger-cap** and **higher market-to-book** stocks. In other words, an important source of liquidity appears to be other mutual funds—especially funds benchmarked to broader or different indexes—rather than only hedge-fund arbitrageurs.
- **Economic takeaway.** Demand curves for stocks are downward sloping, but the size of the index premium depends not only on indexing demand. It also depends on the market’s capacity to absorb that demand through liquidity provision and arbitrage.

2 Data and measurement (key objects)

2.1 Institutional setting: Russell index construction

- **Eligible stocks.** The Russell universe includes U.S. common stocks listed on major U.S. exchanges. ADRs, ADSs, preferred stock, warrants, rights, trust receipts, and stocks priced below \$1 at the end of May are excluded.
- **Index assignment rule.** On the last trading day of May, eligible firms are ranked by market capitalization:
 - ranks 1–1000 → Russell 1000,
 - ranks 1001–3000 → Russell 2000,
 - ranks 1–3000 → Russell 3000E.
- **Reconstitution timing.** Membership is fixed for the following year at the end-of-June reconstitution.
- **Why the upper cutoff matters.** The paper focuses on the **1000 cutoff**, the top of the Russell 2000. Stocks just inside the Russell 2000 get much larger index weights than stocks just outside it in the Russell 1000.
- **Banding policy after 2007.** Starting in 2007, Russell introduced *banding* to reduce turnover. Stocks near the cutoff remained in their previous index unless they moved sufficiently far across a cumulative-market-capitalization band. The paper therefore computes year-specific postbanding cutoffs.

2.2 Observed data sources

- **Russell data.** Annual constituent lists for the Russell 1000 and Russell 2000 are obtained from Russell starting in 1996. Russell 3000E constituent data are available from 2005 onward.
- **Stock-level data.**
 - **CRSP:** prices, returns, trading volume, and shares data.
 - **Compustat:** quarterly shares outstanding and firm fundamentals.
 - **Russell:** float shares and internal data on benchmarked assets.
- **Fund-level data.**
 - **CRSP Mutual Fund Database:** fund performance and fund characteristics.
 - **Thomson-Reuters Mutual Fund Holdings:** fund holdings around reconstitution dates.
- **Sample periods.**
 - Stock-level RD tests: **1996–2012**.
 - Fund-style / liquidity-provider analysis: **1996–2011**.

2.3 Treatment, samples, and running variable

- **Running variable.** The running variable is the stock’s **end-of-May market-capitalization rank** relative to the relevant cutoff.
- **Treatment.** The treatment is **membership in the Russell 2000** after the June reconstitution.
- **Why the RD is fuzzy rather than sharp.** The authors reconstruct rankings using public data. These rankings predict, but do not perfectly reproduce, Russell’s actual assignment because Russell’s exact market-capitalization calculations are not public and because banding affects assignments after 2007.
- **Addition sample.** For the addition effect, the paper studies firms that were in the Russell 1000 before reconstitution and asks whether they just crossed into the Russell 2000.
- **Deletion sample.** For the deletion effect, the paper studies firms that were in the Russell 2000 before reconstitution and asks whether they just crossed into the Russell 1000.
- **Main estimation window.** The preferred specification uses firms within **100 ranks** of the relevant cutoff.

2.4 Key variables

- **Returns.** Monthly raw stock returns.
- **Volume ratio (VR).** A normalized turnover measure:

$$VR_{it} = \frac{V_{it}/V_i}{V_{mt}/V_m},$$

where V_{it} is stock i ’s volume in month t , V_i is its average volume over the previous six months, and the denominator is the analogous market-wide ratio.

- **Short ratio (SR).** Shares sold short divided by shares outstanding.
- **Comovement.** The beta of daily stock returns with respect to Russell 2000 daily returns in a given month, excluding mechanical covariance due to index membership.
- **Institutional ownership (IO).** Quarterly institutional ownership.
- **Validity-check fundamentals.** Repurchase indicator, return on equity (ROE), return on assets (ROA), earnings per share (EPS), assets, cash-to-assets ratio (C/A), interest-coverage ratio (ICR), and float shares.
- **Fund characteristics in the liquidity-provider analysis.**
 - median market capitalization of holdings,
 - median market-to-book of holdings,
 - total net assets,
 - family total net assets,
 - turnover,
 - age,
 - expense ratio,
 - load.

2.5 Key descriptive facts

- **Summary statistics by index (Table 2).**
 - Median market capitalization is about **\$4.21 billion** for Russell 1000 stocks and about **\$0.47 billion** for Russell 2000 stocks.
 - Russell 1000 stocks have higher institutional ownership on average.
 - Russell 2000 stocks have stronger comovement with the Russell 2000 and somewhat more shorting.
- **How much money tracks these indexes (Table 1).**
 - Passive assets tracking the Russell 1000 are typically around **2 to 3.5 times** passive assets tracking the Russell 2000.
 - But when active benchmarking is included, the Russell 2000 often has **more benchmarked dollars** and **many more products** tied to it than the Russell 1000.
 - For example, in 2005 Table 1 reports about **275** products benchmarked to the Russell 2000 versus **48** to the Russell 1000, and about **\$201.4 billion** benchmarked to the Russell 2000 versus **\$90.0 billion** to the Russell 1000.
- **Mechanical demand shift around the cutoff.** In 2002, stocks just left of the 1000 cutoff had roughly **\$5 million** benchmarked to them, while those just right of the cutoff had roughly **\$200 million**. So crossing the cutoff implied about **\$195 million** of extra demand.
- **Relative demand shift.** The stocks around the cutoff had average market capitalization of about **\$1.4 billion**. This implies an average discontinuous demand shock of about **7.3%** of market capitalization.
- **Turnover around the cutoff.**
 - Before banding, around **10%** of Russell 1000 firms moved into the Russell 2000 each year and about **6%** of Russell 2000 firms moved into the Russell 1000.
 - After banding, these turnover rates fell to about **3%** and **2%**, respectively.

2.6 Timing, anticipation, and why the upper cutoff is clean

- **Key dates.**
 - last trading day of May: ranking date,
 - last Friday of June: reconstitution date,
 - main outcome window: May through September.
- **Why front-running is hard at the upper cutoff.**
 - If arbitrageurs buy a stock likely to be added, they can push its price up and move it *out* of the Russell 2000 candidate set.
 - If they short a stock likely to be deleted, they can push its price down and keep it *inside* the Russell 2000.

- **Interpretation.** At the upper cutoff, anticipatory trading is largely self-defeating because trades change the stock's rank and hence its index assignment. This is one reason the upper cutoff provides a particularly clean RD design.

3 Models and empirical specifications (all equations + interpretation)

3.1 Demand shock from moving between indexes

The paper first writes the *net indexing demand* for stock i as

$$\Delta Demand_{it} = w_{i,R2000} BA_{R2000} - w_{i,R1000} BA_{R1000}, \quad (1)$$

where:

- $w_{i,R2000}$ is the stock's weight if it ends up in the Russell 2000,
- $w_{i,R1000}$ is the stock's weight if it stays in the Russell 1000,
- BA denotes benchmarked assets tied to the relevant index.

Using value-weighted indexes, the authors rewrite this as

$$\Delta Demand_{it} = \frac{MV_i}{\sum_{R2000} MV_j} BA_{R2000} - \frac{MV_i}{\sum_{R1000} MV_j} BA_{R1000}. \quad (2)$$

Because the RD compares two otherwise similar stocks around the cutoff, market capitalizations are approximately equal. So the relevant percentage demand shock is

$$\% \Delta Demand = \frac{\Delta Demand_{it}}{MV_i} = \left(\frac{BA_{R2000}}{\sum_{R2000} MV_j} - \frac{BA_{R1000}}{\sum_{R1000} MV_j} \right). \quad (3)$$

Interpretation. The economic force in the paper is very simple: a stock just inside the Russell 2000 receives a much bigger index weight than a very similar stock just outside it in the Russell 1000. Multiplying those weights by benchmarked assets gives the discontinuous demand shock that drives the RD design.

3.2 Fuzzy RD first stage

Let D_{it} denote actual membership in the Russell 2000 after reconstitution, and let τ_{it} be an indicator for whether the reconstructed rank crosses the relevant cutoff c .

The first stage is

$$D_{it} = \alpha_{0l} + \alpha_{1l}(r_{it} - c) + \tau_{it}[\alpha_{0r} + \alpha_{1r}(r_{it} - c)] + \varepsilon_{it}. \quad (4)$$

Interpretation.

- The cutoff indicator τ_{it} is the instrument.
- The design is *fuzzy* because crossing the reconstructed cutoff does not perfectly determine actual Russell membership.

- The coefficient α_{0r} measures how much more likely a stock is to be in the Russell 2000 when it crosses the cutoff.

In the data, this first stage is very strong: around the upper cutoff, crossing the cutoff raises Russell 2000 membership probabilities by about **0.705 to 0.820**, depending on whether one studies additions/deletions and prebanding/postbanding periods.

3.3 Second-stage RD for returns and other outcomes

For an outcome Y_{it} , the second stage is

$$Y_{it} = \beta_{0l} + \beta_{1l}(r_{it} - c) + D_{it}[\beta_{0r} + \beta_{1r}(r_{it} - c)] + e_{it}. \quad (5)$$

Interpretation.

- β_{0r} is the causal RD effect of Russell 2000 membership at the cutoff.
- For returns, this is the addition effect for stocks moving from the Russell 1000 into the Russell 2000, or equivalently the negative of the deletion effect for stocks moving the other way.
- The paper estimates this separately for monthly returns in May, June, July, August, and September.

The preferred specification uses:

- a **bandwidth of 100 ranks**,
- **local linear** functions of rank on each side,
- separate estimation for additions and deletions.

3.4 From price impact to demand elasticity

To relate the return jump to a stock-demand elasticity, the paper uses

$$Elasticity = -\frac{\% \Delta Demand}{\Delta Return}. \quad (6)$$

Interpretation.

- If demand curves were flat, the elasticity would be very large in absolute value (approaching $-\infty$).
- A finite negative elasticity means stock demand is downward sloping.
- Using the estimated demand shock of **7.3%** and the addition return effect of about **5%**, the implied elasticity is about -1.46 .

3.5 Why the preferred running variable is end-of-May rank

The paper argues that using *end-of-June index weights* as the running variable would produce a *nonvalid* RD design for two reasons:

- end-of-June weights depend on Russell’s opaque float adjustments,
- end-of-June weights already incorporate June returns, so they are partly sorting on the outcome itself.

Interpretation. The correct running variable is the *pre-treatment* ranking measured at the end of May, not a post-treatment object such as June index weight.

3.6 Validity and anticipation logic

The RD design requires local randomization around the cutoff. The paper supports this with several arguments:

- **Continuity of market capitalization.** End-of-May market capitalization is smooth through the cutoff.
- **No bunching problem.** Because rankings are ordinal, there is exactly one firm at each rank. So the usual McCrary density test is not informative here.
- **No discontinuities in pre-treatment fundamentals.** Repurchases, profitability, assets, cash ratios, interest coverage, and float are all smooth through the cutoff.
- **Front-running is self-defeating.** Trading to exploit probable addition or deletion changes the stock’s rank and can undo the index change itself.

Interpretation. The paper’s identification comes from a discontinuity in index demand, not from discontinuities in fundamentals or strategic manipulation around the cutoff.

3.7 Time-trend specification

To study how indexing effects evolve over time, the paper interacts the RD treatment effect with a linear time trend t :

$$D_{it} = \alpha_{0l} + \alpha_{1l}(r_{it} - c) + \alpha_{2l}t + \tau_{it}[\alpha_{0r} + \alpha_{1r}(r_{it} - c) + \alpha_{2r}t] + \varepsilon_{it},$$

and

$$Y_{it} = \beta_{0l} + \beta_{1l}(r_{it} - c) + \beta_{2l}t + D_{it}[\beta_{0r} + \beta_{1r}(r_{it} - c) + \beta_{2r}t] + e_{it}.$$

For price-impact regressions, the outcome is often

$$Y_{it} = \frac{Return_{it}}{\% \Delta Demand_{it}}.$$

Interpretation.

- β_{0r} gives the price impact in the base year 1996.
- $\beta_{2r} < 0$ means the price impact of a given demand shock *falls* over time, i.e. demand becomes more elastic.

3.8 Identifying indexers and liquidity providers

The fund-level analysis compares changes in holdings around reconstitution:

- **Indexers:** funds that *increase* holdings of RD addition stocks and *decrease* holdings of RD deletion stocks, each by more than 10%.
- **Liquidity providers:** funds that do the opposite trades.

To remove mechanical price effects, holdings are valued using *pre-reconstitution prices* both before and after reconstitution.

The paper then estimates regressions of the form

$$Indexer_{ft} = \gamma X_{ft} + \lambda_t + u_{ft},$$

where X_{ft} includes log median market capitalization of holdings, log market-to-book, size, turnover, age, expenses, and load.

Interpretation. This lets the authors ask what *types* of mutual funds are most likely to be on the indexing side versus the liquidity-providing side of the reconstitution trade.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard index-event approach

- In the S&P 500 literature, additions are often associated with news coverage, attention, investor recognition, or changing expectations about the firm.
- Using the market as a control group makes it hard to isolate the effect of *forced demand*.
- This is one reason why earlier work often finds a strong addition effect but a weak or nonexistent deletion effect.

Why that matters. If additions and deletions are not studied in a clean way, one cannot tell whether price changes really come from indexing or from confounding information effects.

4.2 What identifies the paper's causal effect

- Stocks just above and just below the Russell 2000 upper cutoff are extremely similar in size.
- Crossing the cutoff creates a sharp jump in post-reconstitution index weights and hence in benchmarked demand.
- Around the cutoff, assignment is as good as random once one conditions on rank relative to the cutoff.

Key identifying variation. Small random changes in end-of-May market capitalization create large changes in index demand with little change in fundamentals.

4.3 Why the fuzzy RD is necessary

- The authors reconstruct end-of-May rankings from public CRSP/Compustat data.
- Russell’s exact ranking procedure is not directly observable, and post-2007 banding further complicates exact prediction.
- So the cutoff indicator is a strong but imperfect predictor of actual Russell 2000 membership.

Interpretation. The instrument is clean, but treatment assignment is measured with some noise; hence the correct design is fuzzy RD rather than sharp RD.

4.4 Why manipulation is unlikely

- End-of-May market capitalization is smooth through the cutoff.
- There are no discontinuities in repurchase behavior or in financial-slack variables.
- There are no breaks in profitability, assets, or float.
- Exact one-firm-per-rank construction eliminates the usual “bunching” concern.

Interpretation. There is little evidence that firms strategically manipulate their capitalization to land in the preferred index.

4.5 Why anticipation does not kill the design

- At the upper cutoff, buying a likely addition candidate can push it out of the Russell 2000; shorting a likely deletion candidate can keep it in.
- This self-defeating feature limits profitable front-running.
- The paper notes that the lower Russell 2000 cutoff (around 3000) is less clean for exactly this reason.

Interpretation. The upper cutoff is especially credible because the reconstitution rule itself blunts anticipatory arbitrage.

4.6 What identifies the time trends and liquidity-provider analysis

- Time trends are identified by repeating the same RD experiment year after year and interacting treatment effects with calendar time.
- Liquidity-provider types are identified by differential pre/post reconstitution changes in holdings of RD addition and deletion stocks.

Interpretation. The paper is not just measuring one average addition effect. It is using the same quasi-experiment repeatedly to study how market structure changed over time.

4.7 Relation to the literature

- The paper contributes to the literature on downward-sloping demand curves for stocks.
- It differs from prior S&P 500 work by providing clean causal estimates of both *addition* and *deletion* effects.
- It also contributes to the literature on indexing and comovement by showing that greater indexing need not imply a larger index premium if arbitrage capacity also rises.

5 Tables and figures

5.1 Figure 1 and Figure 2: the RD design and the mechanical discontinuity

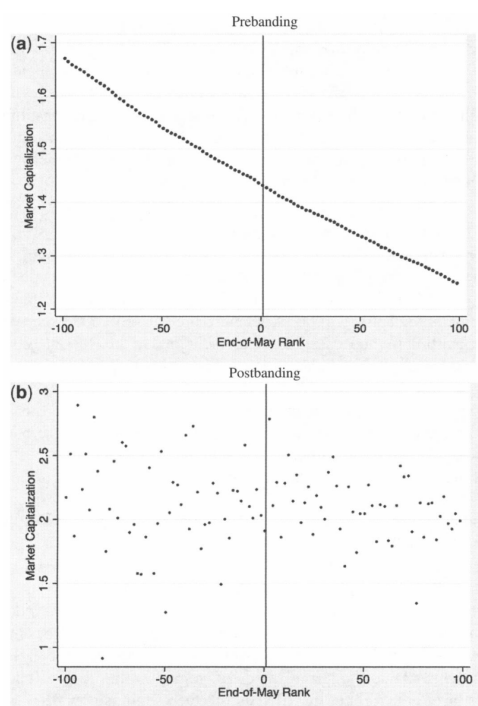


Figure 1
End-of-May market capitalization around upper cutoff
 End-of-May market capitalization is measured in billions of dollars and plotted against end-of-May rankings. Firms that will end up in the Russell 1000 are on the left-hand side of the cutoff and firms that will end up in the Russell 2000 are on the right-hand side. The sample period is 1996–2006 for the prebanding period and 2007–2012 for the postbanding period.

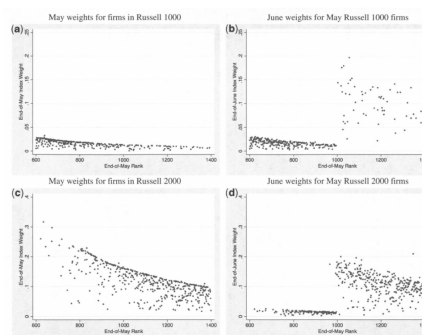


Figure 2
Index weights around upper cutoff in 2002
 Index weights are measured in percent of index (in percentage points) and plotted against end-of-May market capitalization rankings. Panels (a) and (b) use firms that were in the Russell 1000 at the end of May while panels (c) and (d) use firms that were in the Russell 2000 at the end of May. The firms that end up in the Russell 1000 are on the left-hand side of the cutoff. Those that end up in the Russell 2000 are on the right-hand side.

Read:

- Figure 1 is the first identification check. End-of-May market capitalization is smooth through the cutoff both before and after banding. This supports local randomization.
- Figure 2 shows the key mechanical force in the paper: *index weights jump sharply at the cutoff after reconstitution*.
- For stocks near the 1000 cutoff, moving from the Russell 1000 to the Russell 2000 increases weights by roughly a factor of **ten**.

- The same logic works in reverse for deletions, so the paper expects addition and deletion effects to be of similar magnitude.

5.2 Table 1 and Figure 3: benchmarked assets and implied demand shocks

Table 1
Assets benchmarked to indexes

Panel A: Passive assets

	1996	1997	1998	1999	2000	2001	2002	2003
Russell 2000	11.6	7.6	11.0	13.6	18.9	21.5	26.9	24.6
Russell 1000	20.9	20.7	19.0	25.9	17.3	34.0	35.6	37.2
	2004	2005	2006	2007	2008	2009	2010	2011
Russell 2000	38.9	39.2	43.0	51.7	38.5	38.4	56.8	60.1
Russell 1000	84.9	93.3	151.9	175.8	144.8	104.4	137.1	125.8

Panel B: Assets benchmarked

Number of products	2002	2003	2004	2005	2006	2007	2008
S&P 500	1,009	924	919	901	888	824	685
Russell 2000	289	255	264	275	273	511	449
Russell 1000	29	43	43	48	52	52	60
Dollar amount	2002	2003	2004	2005	2006	2007	2008
S&P 500	1,679.8	1,096.9	1,431.8	1,482.9	1,576.7	1,748.6	1,412.1
Russell 2000	198.2	140.7	162.5	201.4	221.1	291.4	263.7
Russell 1000	47.6	37.3	66.9	90.0	146.1	172.7	168.6

Panel A reports the dollar amount of passive assets, in billions, benchmarked to the Russell 1000 and Russell 2000 by year. The data come from Russell's internal unaudited survey of its clients at the end of June. Panel B reports the number of products and dollar amount (in billions) of institutional assets benchmarked to the Russell 2000, Russell 1000, and S&P 500. These numbers are taken from Russell Investment's 2008 U.S. Equity Indexes: Institutional Benchmark Survey. The products surveyed are primarily institutional-oriented mutual funds, separate accounts, and commingled funds at the end of May.

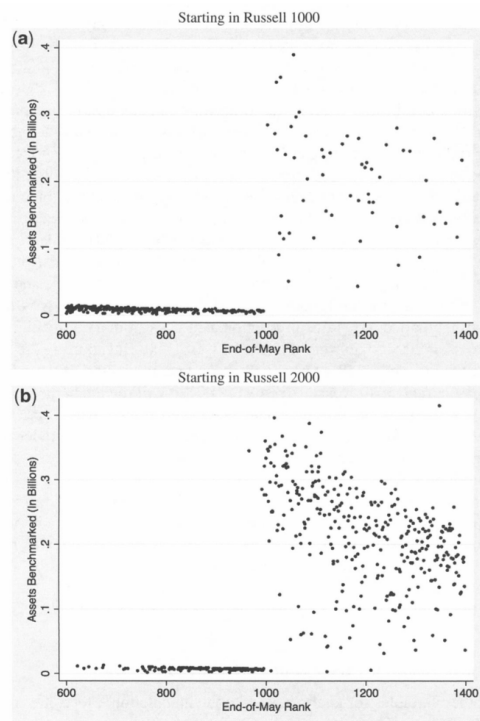


Figure 3
Assets benchmarked to stocks in 2002
Assets benchmarked are measured in billions of dollars and plotted against end-of-May market capitalization rankings. This is calculated by multiplying June index weights by the dollars benchmarked to the relevant index (from Panel B of Table 1) in 2002. Firms on the left side of the cutoff end up in the Russell 1000 in June and those on the right side end up in the Russell 2000.

Read:

- Table 1 shows why the weight discontinuity matters economically, not just mechanically.
- Passive assets following the Russell 1000 are often larger, but when *benchmarking* is included, the Russell 2000 attracts many products and substantial dollars.
- Figure 3 translates this into stock-level demand: around the upper cutoff in 2002, stocks just inside the Russell 2000 had about **\$200 million** benchmarked to them, versus only about **\$5 million** for comparable stocks just outside.
- This produces an implied demand shock of about **7.3% of market capitalization**, which is large enough to generate economically meaningful price effects.

5.3 Table 2 and Table 6: summary statistics and validity checks

Read:

- There are no meaningful effects in May, which is exactly what one wants if the design is valid, because treatment has not yet occurred.
- There are also no meaningful effects in July–September, so the price adjustment is fast and there is little evidence of reversal.
- Figure 4 makes the discontinuity visual: June returns jump at the cutoff for additions and for deletions.
- The symmetry of addition and deletion is one of the strongest pieces of evidence that the paper is identifying a *forced-demand* effect rather than recognition or publicity.

5.5 Table 5: trading volume and institutional ownership

Table 5
Other outcome variables

Addition effect					
	May	Jun	Jul	Aug	Sep
VR	-0.010 (-0.13)	0.478** (3.14)	0.114 (1.44)	0.097 (1.15)	-0.020 (-0.22)
IO		0.031 (0.77)			0.036 (0.89)
<i>N</i>	920	921	918	915	913
Deletion effect					
	May	Jun	Jul	Aug	Sep
VR	0.151 (0.79)	-0.263** (-2.74)	0.050 (0.55)	0.106 (1.55)	0.166 (1.64)
IO		-0.063 (-1.69)			-0.037 (-1.02)
<i>N</i>	1308	1309	1300	1295	1284

The table reports the results of a fuzzy RD design. The following equation is estimated.

$$Y_{it} = \beta_{0t} + \beta_{1t}(r_{it} - c) + D_{it}[\beta_{0t} + \beta_{1t}(r_{it} - c)] + \epsilon_{it}.$$

The independent variable D is an indicator for membership in the Russell 2000 index. An indicator for whether ranking r_{it} is above the cutoff c is used as an instrument for D . We show coefficient estimates of β_{0t} , and t -statistics are reported in parentheses. The bandwidth is 100. The regression identifying the addition effect only uses firms that were in the Russell 1000 at the end of May. VR is volume ratio. IO is institutional ownership and is measured quarterly. The sample period is 1996–2012. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 7
Addition effect over time

	May	Jun	Jul	Aug	Sep
Returns/% Δ Demand	0.243 (0.10)	5.850** (2.76)	-3.791 (-1.37)	1.273 (0.49)	-1.761 (-0.61)
Returns/% Δ Demand $\times t$	0.088 (0.50)	-0.403* (-2.46)	0.386 (1.87)	0.080 (0.39)	0.115 (0.55)
<i>N</i>	1026	1028	1024	1023	1018
VR	-0.026 (-0.23)	0.329* (2.00)	0.055 (0.58)	0.166 (1.55)	0.015 (0.12)
VR $\times t$	0.004 (0.38)	0.023* (2.50)	0.010 (1.25)	-0.009 (-0.99)	-0.006 (-0.64)
<i>N</i>	1048	1049	1051	1047	1047
SR	-0.010 (-1.03)	-0.013 (-1.28)	-0.016 (-1.57)	-0.021* (-2.10)	-0.018 (-1.82)
SR $\times t$	0.001 (1.41)	0.002* (2.18)	0.004*** (3.44)	0.004*** (3.45)	0.004*** (3.59)
<i>N</i>	920	921	918	915	913
Comovement	0.180 (0.95)	0.133 (0.88)	-0.139 (-0.68)	-0.102 (-0.53)	0.145 (0.70)
Comovement $\times t$	-0.007 (-0.59)	-0.027* (-2.28)	0.017 (1.16)	0.037** (2.64)	-0.001 (-0.03)
<i>N</i>	1049	1050	1046	1045	1042

This table reports the results of a fuzzy RD design where the effect of the instrument is allowed to vary linearly with time. The following equation is estimated.

$$Y_{it} = \beta_{0t} + \beta_{1t}(r_{it} - c) + \beta_{2t}t + D_{it}[\beta_{0t} + \beta_{1t}(r_{it} - c) + \beta_{2t}t] + \epsilon_{it}.$$

The independent variable D is an indicator for membership in the Russell 2000 index. An indicator for whether ranking r_{it} is above the cutoff c is used as an instrument for D . For each regression, we report the coefficient estimates β_{0t} , and β_{2t} , and t -statistics are reported in parentheses. The bandwidth is 100. Only firms that were members of the Russell 1000 index at the end of May are used to estimate the addition effect. Returns/% Δ Demand is monthly returns weighted by the inverse demand change. VR is volume ratio. SR is short ratio. Comovement is the beta estimate between daily stock returns and Russell 2000 index returns, computed monthly and excluding mechanical covariance due to index membership. The sample period is 1996–2012. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Read:

- The June **volume-ratio** effect is large:
 - for additions, June *VR* rises by about **0.478**,
 - for deletions, June *VR* falls by about **0.263** for stocks that remain in the index, implying more trading in deleted stocks.
- This is exactly what one expects when indexers rebalance.
- But institutional ownership does *not* change significantly.
- The natural interpretation is that some institutions with more inelastic demand trade with other institutions willing to provide liquidity, rather than all the adjustment showing up as a permanent shift in institutional ownership.

5.6 Table 7, Table 8, and Figure 5: time trends in price impact, trading, shorting, and comovement

Table 8
Deletion effect over time

	May	Jun	Jul	Aug	Sep
Returns/% Δ Demand	0.137 (0.05)	8.661*** (3.91)	-4.106 (-1.44)	-1.460 (-0.53)	1.194 (0.44)
Returns/% Δ Demand $\times t$	-0.108 (-0.44)	-0.608*** (-3.70)	0.455* (2.20)	0.056 (0.26)	-0.091 (-0.45)
N	1473	1472	1461	1454	1447
VR	0.229 (0.88)	-0.057 (-0.51)	-0.034 (-0.33)	0.164 (1.83)	0.093 (0.88)
VR $\times t$	-0.010 (-0.70)	-0.028** (-3.08)	0.011 (1.13)	-0.008 (-1.23)	0.010 (1.15)
N	1543	1545	1542	1532	1523
SR	0.007 (0.71)	0.003 (0.34)	0.002 (0.26)	0.006 (0.59)	0.007 (0.71)
SR $\times t$	-0.000 (-0.64)	-0.000 (-0.27)	0.001 (1.84)	0.001 (1.71)	0.001 (1.28)
N	1308	1309	1300	1295	1284
Comovement	-0.062 (-0.43)	0.321 (1.92)	0.353* (2.24)	-0.083 (-0.48)	0.265 (1.43)
Comovement $\times t$	0.006 (0.56)	-0.000 (-0.03)	0.002 (0.23)	0.034** (2.71)	0.004 (0.32)
N	1540	1540	1532	1521	1514

This table reports the results of a fuzzy RD design where the effect of the instrument is allowed to vary linearly with time. The following equation is estimated.

$$Y_{it} = \beta_{0t} + \beta_{1t}(r_{it} - c) + \beta_{2t}t + D_{it}[\beta_{0r} + \beta_{1r}(r_{it} - c) + \beta_{2r}t] + \epsilon_{it}$$

The independent variable D is an indicator for membership in the Russell 2000 index. An indicator for whether ranking r_{it} is above the cutoff c is used as an instrument for D . For each regression, we report the coefficient estimates β_{0r} and β_{2r} , and t -statistics are reported in parentheses. The bandwidth is 100. Only firms that were members of the Russell 2000 index at the end of May are used to estimate the deletion effect. Returns/% Δ Demand is monthly returns weighted by the inverse demand change. VR is volume ratio. SR is short ratio. Comovement is the beta estimate between daily stock returns and Russell 2000 index returns, computed monthly and excluding mechanical covariance due to index membership. The sample period is 1996–2012. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

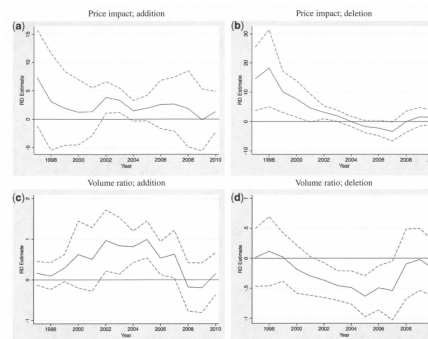


Figure 5
RD estimates over time
For each year, the RD estimate (β_0 in Equation (6)) is computed by pooling data with three surrounding years and estimating the RD effect of index addition or deletion. The data begin one year before and continue two years after the year point. Price impact and volume ratio are measured in June. All regressions use firms with end-of-May ranking within 100 spots of the predicted cutoff. The regression specification is linear in ranking and is allowed to vary on either side of the cutoff. The solid line represents the point estimates of the effect of index addition or deletion. The dashed lines represent the 95% confidence interval.

Read:

- Table 7 shows that the *addition* price impact per unit of demand is positive in the early years but declines over time:
 - base-year June coefficient on $Return/\% \Delta Demand$ is about **5.850**,
 - the time-trend coefficient is about -0.403 .
- Table 8 shows the same for *deletions*:
 - base-year June coefficient about **8.661**,
 - trend about -0.608 .
- The main message is that **demand became more elastic over time**, even though indexing grew.
- Volume effects get stronger over time, consistent with greater rebalancing activity.
- Shorting rises over time after additions, which is consistent with more arbitrage capacity.
- Comovement does *not* show a clear upward trend; if anything, the evidence points away from increasing nonfundamental comovement.
- Figure 5 is helpful because it shows these trends visually. The decline is especially clear for the deletion effect.

5.7 Table 9: who provides liquidity to indexers?

Read:

Table 9
Fund characteristics of indexers and liquidity providers

	All	1996–2003	2004–2011
Log(MV)	−0.0704*** (−4.57)	−0.0508* (−2.50)	−0.0905*** (−3.82)
Log(MtB)	−0.251*** (−7.14)	−0.223*** (−5.47)	−0.363*** (−5.28)
Log(TNA)	0.00316 (0.36)	0.00601 (0.50)	−0.000540 (−0.04)
Log(1 + FAM TNA)	0.00708 (1.43)	0.00366 (0.56)	0.0116 (1.56)
TURN	−0.00116 (−0.13)	0.0193 (1.70)	−0.0135 (−1.36)
AGE	−0.000270 (−0.19)	−0.00138 (−0.82)	0.000963 (0.42)
EXP RATIO	−1.185 (−0.39)	−4.722 (−1.25)	5.470 (1.04)
LOAD	−0.237 (−0.71)	−0.451 (−1.04)	0.105 (0.20)
<i>N</i>	14838	7841	6997
adj. <i>R</i> ²	0.111	0.084	0.159

This table reports the OLS regression results of fund style assignment on fund characteristics. The dependent variable is an indicator variable that equals one (zero) if a fund is flagged as an indexer (liquidity provider). Log(MV) and Log(MtB) are the log of median market capitalization and market-to-book of fund holdings, measured annually using the latest report date from April of year $t-1$ to March of year t . MtB is winsorized by dropping at 99.75% and assumes a five-month lag in book value. Log(TNA), Log(1+ FAM TNA), TURN, AGE, EXP RATIO, and LOAD are a fund's lagged log total net assets (in millions), log fund family total net assets, turnover, age, expense ratio, and total load, all measured monthly. t -statistics are in reported in parentheses. The sample period is 1996–2011. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

- Table 9 compares funds identified as Russell 2000 *indexers* with funds identified as *liquidity providers*.
- The coefficients on $\log(MV)$ and $\log(MtB)$ are negative and statistically significant in the indexer-dummy regression.
- So indexers tend to hold **smaller-cap** and **lower market-to-book** stocks, while liquidity providers tend to hold **larger-cap** and **growth** stocks.
- This suggests that funds benchmarked to other broad or growth-style indexes are important liquidity providers for Russell 2000 index funds.
- The paper's theoretical implication is important: liquidity to indexers need not come only from hedge-fund arbitrageurs. It can also come from other institutions with different benchmark mandates.

6 Short summary

- **Method.** The paper uses a fuzzy regression-discontinuity design around the Russell 2000 upper cutoff to estimate the causal price effects of index membership.
- **Identification.** Stocks just above and just below the cutoff are similar in fundamentals but face a sharp discontinuity in index demand because of value-weighted Russell index construction.
- **Main findings.** Addition to the Russell 2000 raises June returns by about **5%**; deletion lowers them by about **5.4%**. The effects are symmetric, concentrated in June, and accompanied by large trading-volume changes but little change in institutional ownership.
- **Broader result.** Indexing demand matters, but its price impact fell over time as markets developed more arbitrage and liquidity-supply capacity. Mutual funds holding larger and growth-oriented stocks appear to play an important role in providing that liquidity.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a clean causal effect of stock-market indexing on prices using the Russell reconstitution rule.
- It shows that both **additions** and **deletions** matter, and that the effects are economically large and roughly symmetric.
- It also shows that the size of indexing effects changes over time and depends on who stands ready to absorb indexers' trades.

7.2 Economic intuition

- A stock near the upper Russell 2000 cutoff has a downward-sloping demand curve. When it crosses into the Russell 2000, benchmarked funds must buy it, and because substitutes are imperfect, the price rises.
- But the eventual price impact does not depend only on the mechanical demand shock. It also depends on how much liquidity is supplied by other institutions and arbitrageurs.
- Over time, more indexing raised the need for rebalancing, but improved liquidity provision and arbitrage made demand more elastic and muted price effects.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it shows how a carefully designed regression discontinuity can cleanly identify asset-demand effects in financial markets. Substantively, it shows that indexing has real price effects, but those effects are not fixed technological constants. They evolve with market structure, the growth of indexing, and the availability of liquidity providers. The central lesson is that stock demand is downward sloping, yet the *market's ability to absorb benchmark-driven trades* is itself an important equilibrium object.